

PROJECT DELIVERABLE - PHASE 4

Project Development Phase

Implementation Details, Script Artifacts, Test Execution & Deliverables

Platform: ServiceNow PDI dev423661 | Global Scope | September 2026

1. ServiceNow Environment & Master Data Implementation

Development was carried out on ServiceNow Personal Developer Instance dev423661.service-now.com running the Australia release. The custom table u_metro_station_details was created with u_station_name as the display column, populated with six key metro stations:

Station Name Record	Station Code	Role in System	Verification Status
Ameerpet	AMP	Primary transit interchange; base statio..	Verified
Madhapur	MDP	IT corridor terminus; 2-passenger fare t..	Verified
LB Nagar	LBN	East-west terminal station; verified in ..	Verified
Kukatpally	KPT	Major residential hub station; active in..	Verified
Jubilee Hills	JBH	Commercial metro hub; verified in test b..	Verified
Uppal Stadium	UPS	Sports and transit hub; validated in des..	Verified

2. Catalog Item & Client-Side Script Automation

The catalog item 'Book A Metro Ticket' was configured in the Services category of Service Catalog with 8 variables. Two core client-side JavaScript scripts were implemented to drive interactive automation:

- Fare Auto-Calculation (onChange Script): Monitors no_of_passengers. Compares starting_from and going_to sys_ids, multiplies passenger count by route fare, doubles for return journeys, and sets read-only amount fields.
- QR Ticket Generator (onSubmit Script): Intercepts form submission, builds a secure QR API request URL using encodeURIComponent, triggers spModal.open with metro_qr_widget, and allows checkout to proceed.

3. Catalog UI Policy Configuration

Dynamic field visibility is enforced using a Catalog UI Policy without custom script overhead:

Policy Attribute	Configured Setting / Value	Functional Behavior
Policy Name	Show Enter Payment Mode	Identifies UI policy rule in Maintain Items
Condition	mode_of_payment is 'Others'	Evaluates selected value in payment dropdown
On Load	True (Checked)	Applies rule when catalog item initially o..
Reverse on False	True (Checked)	Automatically hides field when user switch..
Target Variable	enter_payment_mode	Single Line Text variable controlled by ac..
Policy Action	Visible = True, Mandatory..	Forces commuter to specify custom payment ..

4. Service Portal Widget (metro_qr_widget)

A dedicated Service Portal widget (metro_qr_widget) was created to render the digital ticket inside spModal:

Widget Architecture

HTML Template: <div class='text-center'><img ng-src='{{c.data.qr_url}}'
class='img-responsive' style='margin: 0 auto;'/><p style='margin-top: 10px;
font-weight: bold;'>Scan this QR code at the metro station gate</p></div> Server
Script: data.qr_url = input.qr_url; (passes payload directly from onSubmit client
script to AngularJS view).

5. Testing & Verification Matrix (TC-01 to TC-15)

Comprehensive end-to-end verification was completed on Service Portal (/sp) with the following logged results:

Test ID	Test Scenario	Input Conditions	Expected vs. Actual Outcome	Result
TC-01	Starting Station P..	Select 'Jubilee Hi..	Station reference field populat..	PASS
TC-02	Destination Statio..	Select 'Uppal Stad..	Destination reference field pop..	PASS
TC-03	Journey Type Radio	Select 'Return jou..	Return option radio selected an..	PASS
TC-04	Passenger Count Dr..	Select '4' passeng..	Dropdown selects 4; triggers on..	PASS
TC-05	Conditional Field ..	Select Mode = 'Oth..	enter_payment_mode becomes visi..	PASS
TC-06	UI Policy Reversal	Switch Mode to 'UPI'	enter_payment_mode immediately ..	PASS
TC-07	Fare Calculation -..	Ameerpet -> Madhap..	Amount shows Rs. 60 (Rs. 30 x 2..	PASS
TC-08	Fare Calculation -..	Ameerpet -> Madhap..	Amount shows Rs. 120 (Rs. 60 x ..	PASS
TC-09	Read-Only Field Gu..	Attempt typing in ..	Fields remain locked and unedit..	PASS
TC-10	Mandatory Field Gu..	Submit blank start..	ServiceNow blocks submission wi..	PASS
TC-11	QR Ticket Generation	Click 'Order Now'	Modal pops up rendering scannab..	PASS
TC-12	Record Persistence	Click 'OK' in QR m..	RITM record created in sc_req_i..	PASS

6. Project Outcomes & Development Summary

All core capabilities of the Metro Ticket Generating System have been built, verified, and documented. The project demonstrates end-to-end proficiency across the ServiceNow platform, covering Service Catalog modeling, GlideForm API client scripting, Catalog UI Policies, custom widget engineering, and third-party REST API integration.